KEY SHEET

MATHEMATICS

1	**B**	2	**D**	3	**A**	4	**B**	5	**ABD**	6	**ABD**
7	**ABC**	8	**136**	9	**7**	10	**6**	11	**5**	12	**31**
13	**23**	14	**55**	15	**66**	16	**18.40**	17	**43.20**		

PHYSICS

18	**C**	19	**C**	20	**C**	21	**A**	22	**BC**	23	**ABC**
24	**ABCD**	25	**8**	26	**21**	27	**8**	28	**2**	29	**18**
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CHEMISTRY

35	**D**	36	**A**	37	**D**	38	**D**	39	**AB**	40	**ABC**
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SOLUTIONS
MATHEMATICS

1. $\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{1}{n^2\tan^{-1}n^2}-\frac{1}{(n+1)^2\tan^{-1}(n+1)^2}\right)=\frac{4}{\pi}$

2. Multiplying complex numbers is equivalent to multiplying their magnitudes and summing their angles.

In order for $\left(\cos(a\pi)+i\sin(b\pi)\right)^4$ to be a real number, then the angle of $\cos(a\pi)+i\sin(b\pi)$ must be a multiple of 45°, so $\cos(a\pi)+i\sin(b\pi)$ satisfies $\cos(a\pi)=0,\sin(b\pi)=0,\cos(a\pi)=\sin(b\pi)$, or $\cos(a\pi)=-\sin(b\pi)$

There are 20 possible values of a and b. $\sin(x\pi)=0$ at $x=\{0,1\}$ and $\cos(x\pi)=0$ at $x=\left\{\frac{1}{2},\frac{3}{2}\right\}$. The probability of $\sin(b\pi)=0$ or $\cos(a\pi)=0$ is $\frac{1}{10}+\frac{1}{10}-\frac{1}{100}$ (we over counted the case where $\sin=\cos=0$)

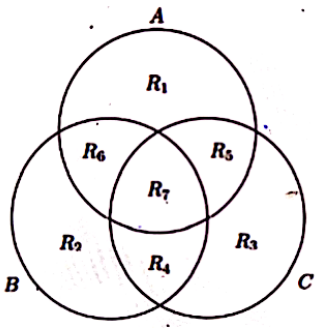
We also consider the case where $\sin(b\pi)=\pm\cos(a\pi)$. This only happens when $a=\{0,1\},b=\left\{\frac{1}{2},\frac{3}{2}\right\}$ or $a,b=\left\{\frac{1}{4},\frac{3}{4},\frac{5}{4},\frac{7}{4}\right\}$. The probability is $\frac{1}{100}+\frac{1}{25}$.

The total probability is $\frac{1}{5}+\frac{1}{25}=\frac{6}{25}$

3. $\frac{1}{7}\times\frac{1}{4}\left(\frac{3}{6}+\frac{2}{6}+\frac{2}{6}+\frac{2}{6}\right)+\frac{1}{4}.\frac{1}{7}\left(\frac{3}{6}+\frac{2}{6}+\frac{2}{6}+\frac{2}{6}+\frac{2}{6}+\frac{2}{6}+\frac{2}{6}\right)$

$\frac{1}{7}\times\frac{1}{4}\times\left(\frac{9}{6}+\frac{15}{6}\right)=\frac{1}{7}$

4.



Consider $N=|R_7|=|A\cap B\cap C|$

Case 1: N = 2

$|R_4|+N=|R_5|+N=|R_6|+N=2$

$|R_4|=|R_5|=|R_6|=0$

$|R_1|+|R_5|+|R_6|+N=4$, so $|R_1|=2$. Similarly, $|R_2|=|R_3|=2$

Number of ways $=\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}=2520$ ways

Case 2: N = 1

$|R_4|+N=|R_5|+N=|R_6|+N=2$

$|R_4|=|R_5|=|R_6|=1$

$|R_1|+|R_5|+|R_6|+N=4$, so $|R_1|=1$. Similarly, $|R_2|=|R_3|=1$

Number of ways $=\binom{8}{1}\binom{7}{1}\binom{6}{1}.....\binom{2}{1}=40320$ ways

Case 3: N = 0

$|R_4|+N=|R_5|+N=|R_6|+N=2$

$|R_4|=|R_5|=|R_6|=2$

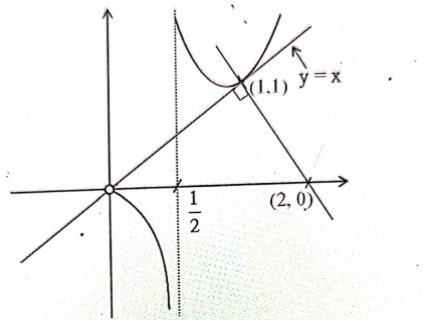
$|R_1|+|R_5|+|R_6|+N=4$,

$|R_1|=0$. Similarly, $|R_2|=|R_3|=0$

Number of ways $=\binom{8}{2}\binom{6}{2}\binom{4}{2}=2520$

Total number of ways = 2520 + 40320 + 2520 = 45360

5.



$\dfrac{d\left(x^3/y\right)}{1+x^3/y}=\dfrac{dx}{x}\Rightarrow \ln\left(1+x^3/y\right)=\ln x+\ln c$

$1+\dfrac{x^3}{y}=cx$

$(1,1)\Rightarrow c=2\Rightarrow 1+\dfrac{x^3}{y}=2x\Rightarrow y=\dfrac{x^3}{2x-1}$

Tangent as (1,1) $\quad y-1=1(x-1)\Rightarrow y=x$

Normal at (1, 1) $\quad y-1=-(x-1)\Rightarrow x+y=2$

6. $f''(x)=f(x).(\tan x-1)\left(\tan^3 x-\tan^2 x+3\tan x+1\right)$

$g(x)=\tan^3 x-\tan^2 x+3\tan x+1=0$ has exactly one root in $x\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ which lies in

$x\in\left(-\dfrac{\pi}{4},0\right)$ So, $f''(x)=0$ at $x=\dfrac{\pi}{4}$ and k where $k\in\left(-\dfrac{\pi}{4},0\right)$

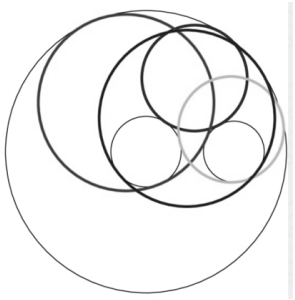
7. $\dfrac{n^4+3n^2+10n+10}{2^n\left(n^4+4\right)}=\dfrac{1}{2^n}+\dfrac{3n^2+10n+6}{2^n\left(n^2+2n+2\right)\left(n^2-2n+2\right)}$

$=\dfrac{1}{2^n}+\dfrac{4n^2+8n+8-n^2+2n-2}{2^n\left(n^2+2n+2\right)\left(n^2-2n+2\right)}=\dfrac{1}{2^n}+\dfrac{1}{2^{n-2}\left(n^2-2n+2\right)}-\dfrac{1}{2^{n-2}\left(n^2+2n+2\right)}$

$\therefore\sum_{n=2}^{\infty}\dfrac{n^4+3n^2+10n+10}{2^n\left(n^4+4\right)}=\sum_{n=2}^{\infty}\dfrac{1}{2^n}+\sum_{n=2}^{\infty}\left(\dfrac{1}{2^{n-2}\left(n^2-2n+2\right)}-\dfrac{1}{2^n\left(n^2+2n+2\right)}\right)$

$=\dfrac{1}{2}+\dfrac{6}{10}=\dfrac{11}{10}$

8.



$4\left(5^2\pi+3^2\pi\right)=136\pi$

9. Let $2^{17}=x$ given equation is $\dfrac{x^{17}+1}{x+1}=x^{16}-x^{15}+x^{14}+\ldots-x^1+x^0$

Now, consider $\left(x^{16}-x^{15}\right)=x^{15}\left(x-1\right)=x^{15}\left(2^{17}-1\right)$

Since, $2^{17}-1=2^{16}+2^{15}+\ldots+1\Rightarrow$ Total 17 terms

So, there are 8 such pair like $x^{14}-x^{13}$ and so on

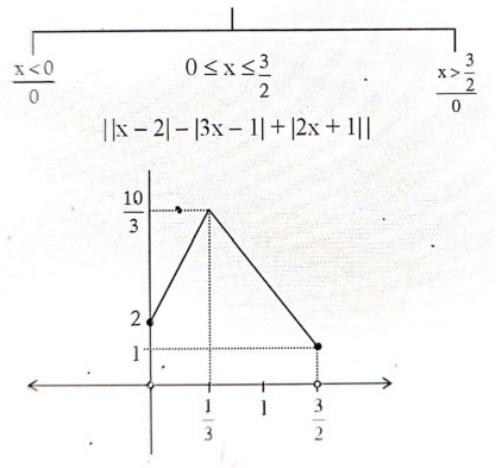
So, total is $17\times8+1=137$, 1 is for x^o

10. $\alpha^7=3$

$x^7-21\left[x^2\left(x+1\right)+x^2+x+1\right]+15$

$=3\left(1+\alpha\right)^7-21\left[\left(\alpha^4+2\alpha^3+\alpha2\right)\left(\alpha^2+\alpha+1\right)+\alpha^4+2\alpha^3+\alpha^2+\alpha^2+\alpha+1\right]+15=6$

11. $h\left(x\right)=\left\|x-2\right|-\left|3x-1\right|+\left|2x+1\right\|\left[\left|-x^2+\dfrac{3}{2}x+1\right|\right]$



12. The key is to count backwards. First choose the pair which you pick on Wednesday in 5 ways. Then there are four pairs of socks for ou to pick a pair of on Tuesay and don't want to pick a pair. Since there are 4 pairs, the number of ways to do this is $\binom{8}{2}-4$. Then, there are two pairs and two nonmatching socks for you to pick from on Monday a total of 6 socks. Since you don't want to pick a pair, the number of ways to do this $\binom{6}{2}-2$. Thus the answer is

$$\frac{(5)\left(\binom{8}{2}-4\right)\left(\binom{6}{2}-2\right)}{\binom{10}{2}\binom{8}{2}\binom{6}{2}}=\frac{26}{315}$$

$$\frac{m+n}{11}=\frac{315+26}{11}=\frac{341}{11}=31$$

13. Let the position vectors of A, B and C be $\vec{O},\vec{b}$ and $\vec{c}$ respectively.

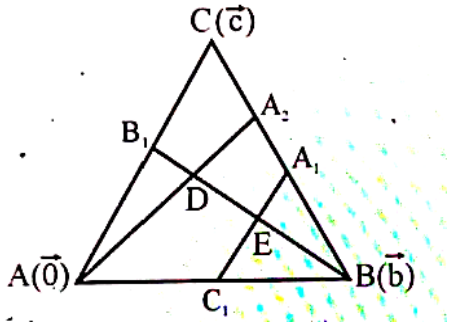
We have, $\overrightarrow{AC_1}=\frac{\vec{b}}{2},\overrightarrow{AB_1}=\frac{\vec{c}}{2},\overrightarrow{AA_1}=\frac{\vec{b}+\vec{c}}{2},\overrightarrow{AA_1}=\frac{3\vec{c}+\vec{b}}{4}$

Equation of lines BB_1, AA_2 and C_1A_1 are respectively

$$\vec{r}=\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)$$

$$\vec{r}=\lambda_2\frac{3\vec{c}+\vec{b}}{4} \text{ and } \vec{r}=\frac{\vec{b}}{2}+\lambda_3\left(\frac{\vec{c}}{2}\right)$$

For point D, we have $\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)=\lambda_2\left(\frac{3\vec{c}+\vec{b}}{4}\right)$



$$\Rightarrow\vec{b}\left(1-\lambda_1-\frac{\lambda_2}{4}\right)+\frac{\vec{c}}{4}\left(2\lambda_1-3\lambda_2\right)=\vec{0}\Rightarrow\lambda_1=\frac{6}{7},\lambda_2=\frac{4}{7}\Rightarrow\overrightarrow{AD}=\frac{3\vec{c}+\vec{b}}{7}$$

For point E we have $\vec{b}+\lambda_1\left(\frac{\vec{c}}{2}-\vec{b}\right)=\frac{\vec{b}}{2}+\frac{\lambda_3}{2}\vec{c}\Rightarrow\lambda_1=\lambda_3=\frac{1}{2}\Rightarrow=\frac{2\vec{b}+\vec{c}}{4}$

Now, $\overrightarrow{EA_1}=\frac{3\vec{c}+\vec{b}-2\vec{b}-\vec{b}}{4}=\frac{2\vec{c}-\vec{b}}{4}$,

$$\overrightarrow{DA_1} = \frac{\vec{b}+\vec{c}}{2} - \frac{3\vec{c}+\vec{b}}{7} = \frac{5\vec{b}+\vec{c}}{14}$$

Area of the quadrilateral $EA_1A_2D = \frac{1}{2}\left|\overrightarrow{EA_2} \times \overrightarrow{DA_1}\right|$

$$= \frac{1}{112}\left|\left(2\vec{c}-\vec{b}\right)\times\left(5\vec{b}+\vec{c}\right)\right| = \frac{1}{112}\left|10\vec{c}\times\vec{b} - \vec{b}\times\vec{c}\right|$$

$$= \frac{11}{112}\left|\vec{c}\times\vec{b}\right| = \frac{11}{56}\times\frac{1}{2}\left|\vec{c}\times\vec{b}\right| = \frac{11}{56} \text{ (area of } \Delta ABC)$$

Thus required ratio of is $\frac{11}{56}$

14, 15

$$\therefore L = \lim_{n\to\infty} 6\left(1+\frac{1}{3}+\frac{1}{3^2}+\ldots\ldots+\frac{1}{3^{n-1}}\right) = 6^3.\frac{3}{2} = 9$$

$$M = Tr(2B_1) = 2\sum_{i=1}^{3}\frac{2i}{2^2} = 1+2+3$$

$$Tr\left(2^2B_2\right) = 2^2\sum_{i=1}^{3}\frac{2i}{2^2} = \frac{1+2+3}{2}$$

$$f(x) = 1-x^3,\ g_m(x) = \frac{5}{\frac{1}{1-x^3}+\frac{1}{1-(2x)^3}+\ldots.+\frac{1}{1-(5x)^3}}$$

$$= \lim_{x\to 0}\frac{\frac{-(-3x^2)}{(1-x^3)^2}+\frac{(-24^8x^2)}{(1-(2x)^3)^2}-\frac{(81^{27}x^2)}{(1-(3x)^3)^2}+\frac{192^{64}x^2}{(1-(4x)^3)^2}+\frac{375^{125}x^2}{(1-(5x)^3)^2}}{3x^2}\cdot\frac{1}{5}$$

$$\therefore 100 - R = 55$$

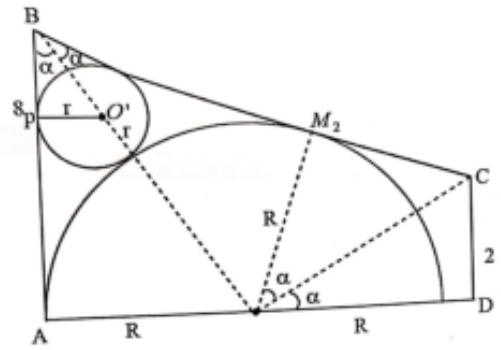
$$L+M+R = 9+12+45 = 66$$

$$Tr(3A_1) = \sum_{i=1}^{3}3\frac{3i}{3^2} = 1+2+3$$

$$Tr\left(3^2A_2\right) = \sum_{i=1}^{3}3^2\frac{3i}{3^4} = \frac{1+2+3}{3}$$

16, 17

$\Delta BAO, \Delta OMC$ are similar



$$\therefore \frac{8}{R} = \frac{R}{2} \Rightarrow R = 4$$

$$\tan\alpha = \frac{4}{8} = \frac{r}{BP} \Rightarrow BP = 2r \ \therefore \left(BO^1\right)^2 = r^2 + (2r)^2 = 5r^2$$

$$BO' = \sqrt{5}r$$

$$OB = \sqrt{4^2+8^2} = 4\sqrt{5}$$

But $OB = BO^1 + r + R = \sqrt{5}r + r + 4$

$$\therefore \left(\sqrt{5}+1\right)r+4=4\sqrt{5} \Rightarrow r=4\frac{\left(\sqrt{5}-1\right)}{\sqrt{5}+1} \Rightarrow r=6-2\sqrt{5}$$

PHYSICS

18. C

Sol: The maximum speed will be acquired by each block in equilibrium.

Let the spring is stretched by x

When the system is in equilibrium,

T = 2mg …. (i)

$T = kx + \mu mg$ ….. (ii)

Solving equation (i) and (ii), we get

$2mg = kx + \mu mg$

$$kx = 2mg - \frac{mg}{2} \Rightarrow x = \frac{3mg}{2k}$$

Now using work energy theorem for the system

$$2mgx - \mu mgx = \frac{1}{2}kx^2 + \frac{1}{2}\left(m+2m\right)v^2$$

$$\left(2mg - \mu mg\right)x = \frac{kx^2}{2} + \frac{1}{2} \times 3mv^2$$

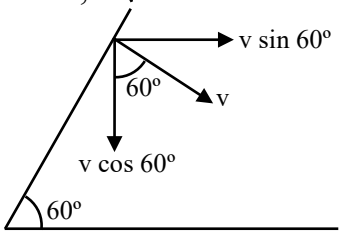
$$kx^2 = \frac{kx^2}{2} + \frac{3mv^2}{2}$$

$$v = \frac{g}{2}\sqrt{\frac{3m}{k}}$$

19. C

Sol. Let v be the velocity at the time of collision.

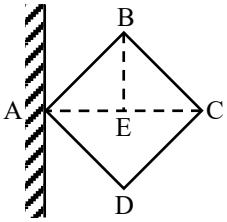
Then, $u\sqrt{2}\cos 45^\circ = v \sin 60^\circ$



$$(u\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = \frac{\sqrt{3}v}{2} \qquad \therefore v = \frac{2}{\sqrt{3}}u$$

20. [C]

Sol. Let AC = x and BE = y



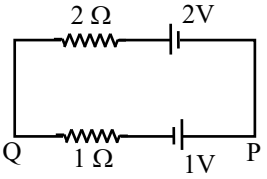
Then, $BE^2 + AE^2 = l^2$ or $y^2 + \left(\frac{x}{2}\right)^2 = l^2$

$\therefore 2y\left(\frac{dy}{dt}\right)+\frac{x}{2}.\frac{dx}{dt}=0$ $\therefore \left(-\frac{dy}{dt}\right)=\frac{1}{2}\left(\frac{x}{2y}\right).\frac{dx}{dt}$

x = 2y, when the rhombus is a square. Hence, $v_B=\frac{1}{2}$ $v_c=\frac{v}{2}$

21. [A]

Sol.



$E_{net}=\frac{E_1 r_2 - E_2 r_1}{r_1+r_2}=\frac{2-2}{2+1}=0$

22. AD

Sol: (i) $I_{max}=\frac{E}{R_2}$

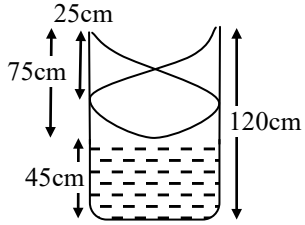
$I_{max}=\left(\frac{\Delta E}{E}+\frac{\Delta R_2}{R_2}\right)I$

(ii) $\frac{\Delta Z}{Z}=\frac{\Delta R_1}{R_1}+\frac{\Delta R_2}{R_2}-\frac{\left(\Delta R_1+\Delta R_2\right)}{R_1+R_2}+\frac{\Delta C}{C}$ $=\frac{R_2^2\Delta R_1+R_1^2\Delta R_2}{R_1R_2\left(R_1+R_2\right)}+\frac{\Delta C}{C}=0.25$

23. **[A,B,C]**

Sol: As $V=v\lambda$

$\lambda=\frac{V}{v}=\frac{340}{340}=1m$



first Resonance length $R_1=\frac{\lambda}{4}=\frac{1}{4}m=25$ cm $\therefore R_2=\frac{3\lambda}{4}=\frac{3}{4}m=75$ cm

$\therefore R_3=\frac{5\lambda}{4}=\frac{5}{4}m=125$ cm

i.e. third resonance does not establish now H_2O is poured

$\therefore$ minimum length of H_2O column to have the resonance = 45 cm

$\therefore$ Distance between two successive nodes = $\frac{\lambda}{2}=\frac{1}{2}m=50$ cm

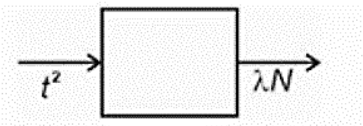
and maximum length of H_2O column to create resonance i.e. 120 – 25 = 95 cm.

24. [A,B,C,D]

Sol:

25. 8

Sol:



$\frac{dN}{dt}=t^2-\lambda N$

for minimum value $\frac{d^2N}{dt^2} = 0 \Rightarrow 2t - \lambda\frac{dN}{dt}$

$\Rightarrow 2t - \lambda\left(t^2 - \lambda N\right) = 0 \Rightarrow 2t - \lambda t^2 + \lambda^2 N = 0$

$\lambda^2 N = \lambda t^2 - 2t$

$N = \left(\frac{\lambda t^2 - 2t}{\lambda^2}\right)$ at t = 4 sec

$N = \frac{16\lambda - 8}{\lambda^2} = \frac{8(2\lambda - 1)}{\lambda^2}$

26. 21

Sol: The torque due to the viscous liquid on the top of the solid cylinder

$\tau_1 = \int_0^R \eta 2\pi r dr \frac{\omega r}{h} r$

$\tau_1 = 2\pi\eta\frac{\omega}{h}\int_0^R r^3 dr$

$\tau_1 = 2\pi\eta\frac{\omega}{h}\frac{R^4}{4} = \frac{\pi\eta\omega R^4}{2h}$ ….. (i)

The torque due to the viscous liquid on the curved surface of the solid cylinder

$\tau_2 = \eta 2\pi Rl\frac{\omega R}{h} R$

$\tau_2 = 2\pi\eta R \times 5R\frac{\omega R^2}{h}$

$\tau_2 = 10\pi\eta\frac{\omega R^4}{h}$ ….. (ii)

Hence the net torque required

$\tau = \tau_1 + \tau_2$

$\tau = \left(\frac{\pi\eta\omega R^4}{2h}\right) + \left(\frac{10\pi\eta\omega R^4}{h}\right)$

$\tau = \frac{21\pi\eta\omega R^4}{2h}$

Hence, K = 21

27. [8]

Sol: At resonance reactance = 0

$I = \frac{V}{R} = \frac{60}{120} = \frac{1}{2}$ Amp

$V_L = I \times X_L = I \times \omega L$

$L = \frac{V_L}{I\omega}$ ……(i)

$\omega_0 = \frac{1}{\sqrt{LC}}$

$C = \frac{1}{L\omega_0^2}$ …….(ii)

Calculate L & C from (1) & (2) current will lag the applied voltage by 45°

if $\tan 45° = \frac{\omega L - \frac{1}{\omega c}}{R}$

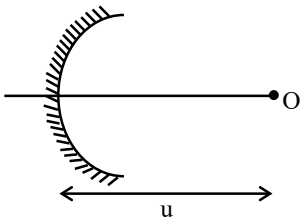
Solve for ω

$\omega = 8 \times 10^5$ rad/sec

28. 2

Sol: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{f} - \frac{1}{u} = \frac{1}{v}$



$v = \frac{fu}{u-f}$

$m = \frac{-v}{u} = \frac{f}{f-u}$

$m_1 \times m_2 = 4$

$\frac{f}{f-25} \times \frac{f}{f-40} = 4$

$f^2 = 4\,[f^2 - 65 + 1000] = 0$

$3f^2 - 260f + 4000 = 0$

$\therefore f = 20$

29. 18

Sol. Fringe width, $w = \frac{\lambda D}{d} \propto \lambda$ wavelength is decreasing from 600 nm to 400 nm so fringe width is also decreasing by a factor of $\frac{4}{6}$ or $\frac{2}{3}$. So no. of fringes will increase by a factor of $\frac{3}{2}$.

30. [154]

Sol. By Mosely law $\frac{1}{\lambda} \propto (Z-1)^2$

$\frac{\lambda_2}{193} = \frac{(26-1)^2}{(29-1)^2}$

$\lambda_2 = \frac{193 \times 25^2}{28^2}$

$\lambda_2 = 153.8\,pm \simeq 154\,pm$

31. 21.00

Sol: The force exerted by the gas on the piston at the moment when compression of the spring is x, is given by $F = P_0A + kx$

where P_0 = 100 k Pa = atomospheric pressure

$A = 20cm^2$ = Area of cross-section of the piston and k = 200 N/m = spring constant.

Hence, work done by the gas as the piston moves through

$l = 10cm$, is given by:

$$W = \int_0^l F dx = \int_0^l \left(P_0 A + kx\right) dx = P_0 Al + \frac{1}{2} kl^2$$

$= (100 \times 103\ \text{N/m}^2) \times (20 - 10^{-4}\text{m}^2) \times (10 \times 10^{-2}\text{m})$

$+ \frac{1}{2} \times (200\text{N/m}) \times (100 \times 10^{-4}\ \text{m}^2)$

$= 20\ \text{J} + 1\ \text{J} = 21\ \text{J}$

32. 793.00

Sol: Final temperature of gas = 331 K

The internal energy is : $U = \left(\frac{3}{2}\right)nRT$

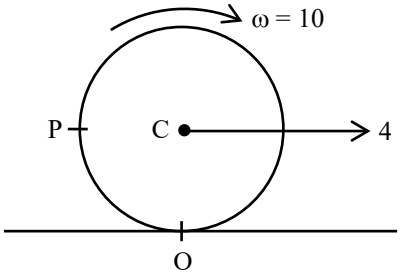
$$\therefore dU = \left(\frac{3}{2}\right) nR\Delta T = 1.5 \times (2.0\text{mol}) \times (8.3\ \text{J/mol–K}) \times (31\text{K}) = 772\ \text{J}$$

Hence, according to first law of thermodynamics,

$\Delta Q + \Delta U + \Delta W = 772J + 21J = 793J$

33. 2.00

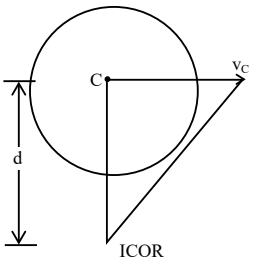
Sol.



$v_0 = v_C - \omega R = 4 - 10 \times 0.2 = 2\ \text{m/s}$

34. 0.20

Sol.



$\omega d = v_C$

$10 \times d = 4$

d = 0.4 m below C i.e. 0.2 m below O.

CHEMISTRY

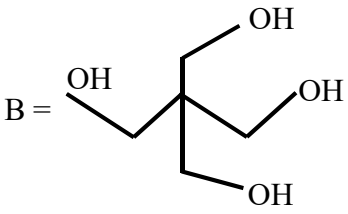
35.

Ans: D

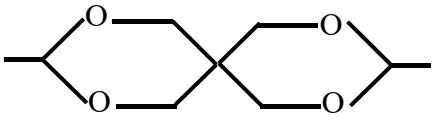
Sol: $M(AA)_2$ can't show G.I

36.

Ans: A

B = 

Sol: $X = CH_3CHO \qquad A = HCOO^{\Theta}Na^{+}$

Y = 

37.

Ans: D

Sol: Aniline is stabilized by resonance.

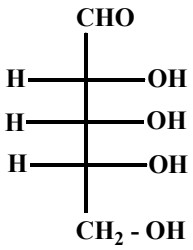
38.

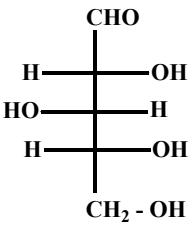
Ans: D

Sol: $\frac{a}{\sqrt{2}} = 5A^{\circ} \Rightarrow a = 5\sqrt{2} = 7.07A^{\circ}$

39.

Ans: AB

Sol: X can be 

and 

40.

Ans: ABC

Sol: $A = PbSO_4,\ B = \left[Pb(CH_3COO)_4\right]^{+2},\ C = (NH_4)_2SO_4$

41.

Ans: A,B,D

Sol: $A \rightarrow 3B$

t = 0	2	1
t = 90	2-x	1+3x

$$\frac{2-x}{1+3x} = \frac{1}{25} \Rightarrow x = \frac{7}{4}$$

$$A_t = 2 - x = \frac{1}{4} = \frac{A_o}{8} \Rightarrow t = 90\,\text{min} = 3t_{1/2} \qquad \therefore t_{1/2} = 30\,\text{min}$$

$B_t = 1 + 3x = \frac{25}{4}$

$n_O = 3 moles \qquad V_t = 12L$

$n_t = 3 + 2x = \frac{26}{4} moles \qquad V_t = ?$

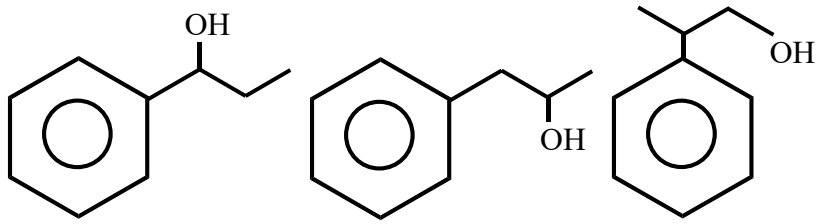
$\frac{26}{4} \times \frac{12}{3} = 26L \therefore [A_t] = \frac{1/4}{26} = \frac{1}{104} M$

$[B_t] = \frac{25/4}{26} = \frac{25}{104} M$

42.

Ans: 6

Sol:



43.

Ans: 8

Sol: Anion of x is NO_3^-

Cation of x is Cu^{+2} $\qquad \therefore x = Cu(NO_3)_2 \quad z = Cu_2I_2$

44.

Ans: 6

Sol: a, b, c, e, f, h

45.

Ans: 3

Sol: If no S – P mixing is there e^- filling order is (b, e, f)

$\sigma s, \sigma_{1s}^*, \sigma_{2s}, \sigma_{2s}^*, \sigma_{2pz}, \pi 2p_y \approx \pi 2p_x ....$

46.

Ans: 3

Sol:

$P\left(V - \frac{W}{M} b\right) = \frac{W}{M} RT$

$P(M - db) = dRT$

$P(68 - 8 \times 0.5) = 8 \times 0.08 \times 300 \Rightarrow P = 3$

47.

Ans: 6

Sol: $P_T = P_A^\circ X_A + P_B^\circ (1 - X_A) = P_B^\circ + (P_A^\circ - P_B^\circ) X_A$

On comparing, we get $P_B^\circ = 100,\ P_A^\circ - P_B^\circ = 200 \Rightarrow P_A^\circ = 300$

Composition of condensate $\frac{YA}{YB} = \frac{300}{100} \times \frac{2/5}{3/5} = \frac{2}{1}$

Vapour above condensate : $\frac{ZA}{ZB}=\frac{300}{100}$ $\frac{YA}{YB}=\frac{300}{100}\times\frac{2}{1}$

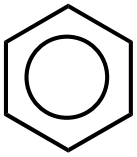
48.

Ans: 38.7 to 38.9

Sol:

49.

Ans: 47.94

Sol: $A = CH \equiv CH$ B =  C = 666, $D = CH_3COCH_3$

1) $C=\frac{1}{2}\times 1\times\frac{1}{3}\times\frac{8}{10}\times 1\times 291=38.8$

2) 1 mole $\rightarrow$ 429.4 gms Br_2

0.1 mole $\rightarrow 429.4\times\frac{1}{10}=42.94$

[Consider At. Wt C = 12, H = 1, Cl = 35.5 and Br = 79.9]

50.

Ans: 79 – 80

Sol: n - factor = $\left[3-\frac{200}{96}\right]\times 0.96=0.88$

Mol.wt = $0.96\times 56+16=69.76$ $\therefore Eq.wt=\frac{69.76}{0.88}=79.27$

51.

Ans: 91.5 – 91.8

Sol: $Fe_{0.96}O \Rightarrow 96'Fe'$ for 100 'O'

Let x = Fe^{+2} & rest Fe^{+3}

Then total charge = $0 \Rightarrow 2x+3(96-x)=200$

$\Rightarrow x=88$

$\therefore \%Fe^{+2}=\frac{88}{96}\times 100=91.67\%$